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Education

Sahyadri College of Engineering and Management, Mangaluru

Expected 2027

B.E. in Computer Science (Artificial Intelligence & Machine Learning)

CGPA: 9.46

Research and Innovation

Multimodal Feature Level Fusion for Arecanut Quality Assessment Article in preparation

Preprocessed and augmented a multimodal RGB and X-ray dataset for arecanut quality assessment, training multiple CNN architectures to achieve 99.80% accuracy. Built an end-to-end Python pipeline using PyTorch and TensorFlow/Keras with GPU acceleration, modular training workflows, and systematic hyperparameter tuning; evaluated models via cross-validation, confusion matrix, precision-recall, and ROC-AUC.

Projects

RxSignal — Drug Interaction Risk Intelligence Platform — Python, XGBoost, PyTorch, FastAPI, React

- Built an end-to-end drug safety platform ingesting 1.06B FDA adverse event records (FAERS 2025) with a chunked DuckDB pipeline, computing PRR/ROR signals across 43.8M drug-event triples.
- Trained XGBoost (90.6% accuracy, 99.35% recall) and a GraphSAGE GNN on a 13,160-node drug graph; served predictions via FastAPI and React in ~1ms using pre-computed embeddings.

Orchestration Agent — Multi-Agent AI System, TDD, Gemini/OpenAI

- Built an autonomous multi-agent system that plans, generates, tests and self-corrects code using a Test-Driven Development (Plan → Code → Test → Fix → Verify) workflow.
- Designed specialized agents (Planner, Coder, Tester, Fixer, Judge) with Pydantic schemas, dual-LLM support (Gemini/OpenAI), rate-limit handling and sandboxed execution with logging.

EEG-Based Early Detection of Neurodegenerative Disorders — Python, TensorFlow, React, FastAPI

- Built a CNN-BiLSTM and GCN deep learning pipeline on 19-channel EEG signals to classify Alzheimer's, FTD and Cognitively Normal with 96.74
- Deployed a FastAPI backend for EEG preprocessing and a React frontend with interactive brain connectivity graph visualization.

Skills

Languages: Python, SQL, Java, C/C++, HTML/CSS, JavaScript.

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, React, Node.js, Express.js.

Databases & Stack: MongoDB, MERN Stack.

AI, ML & Data Science: Machine Learning, Deep Learning, Data Science, Data Analysis, Data Visualization, Feature Engineering, Model Training & Evaluation, Predictive Modeling.

Tools & Platforms: Git, GitHub, Jupyter, Vercel, MongoDB Atlas, Postman, n8n, Microsoft Power BI, Figma, Miro, Canva.

Soft Skills: Strategic Thinking, Problem-Solving, Collaboration, Leadership, Teaching & Mentoring.

Contributions: Solved 140+ LeetCode problems and actively contributed to open source projects.

Achievements

Chair, IEEE Computational Intelligence Society

2025 – Present

Manage chapter activities, coordinate events, and promote research and learning initiatives in computational intelligence. Facilitate student participation in competitions at institutional and external levels.